PRERNA MAHESHBHAI PATIL

M.C.A. - Computer Applications

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BRIEF SUMMARY

I am an MCA student specializing in Artificial Intelligence & Data Analytics at DY Patil International University, passionate about Data Analytics and Machine Learning. I have worked on Blinkit Grocery Data Analysis, Car Price Prediction, and Student Performance - Exploratory Data Analysis (EDA), where I leveraged data processing, visualization, and analytical techniques to extract valuable insights. Proficient in Python, SQL, Excel, and Power BI, I have experience in building interactive dashboards, performing data-driven analysis, and uncovering patterns to support business decisions. I am continuously refining my analytical and problem-solving skills to make meaningful contributions to the field of Data Analytics, using data to drive insights and improve decision-making.

KEY EXPERTISE

Data Analysis | Data Visualization | Data Cleaning | Machine Learning | Python | Pandas | NumPy | Matplotlib | seaborn | SQL | scikit-learn | Power BI

EDUCATION

D. Y. Patil International University, Akurdi — 2024 - 2026

M.C.A. - Computer Applications | CGPA: **9.42** / **10**

Veer Narmad South Gujarat University — 2021 - 2024

B.C.A. - Computer Applications - Computer Applications (Full Time) | CGPA : **7.75** / **10**

Laxmiben DahyaBhai Highschool, Surat — 2021

12th | GSEB | Percentage: **81.33** / **100**

Laxmiben DahyaBhai Highschool, Surat — 2019

10th | GSEB | Percentage: **77** / **100**

PROJECTS

Car Price Prediction

Team Size: 2

Key Skills:

Python | NumPy | Scikit-Learn | Matplotlib | Pandas | seaborn | Data Processing | Data Cleaning | EDA | Encod

Project Link: https://github.com/prerna-patil26/car-price-prediction

Project Overview:
This project focuses on predicting car prices based on key attributes such as brand, model, year, mileage, fuel type, and transmission. Using Machine Learning models, it provides accurate price estimations, helping car buyers, sellers, and dealerships make informed decisions.

My Role:
As the sole developer of this project, I handled the entire workflow, including:

Data Collection & Cleaning – Processed raw data, handled missing values, and performed feature engineering.

Exploratory Data Analysis (EDA) – Visualized trends, correlations, and patterns using Matplotlib and Seaborn.

Model Development & Evaluation – Implemented and compared multiple machine learning models, including Linear Regression, Decision Tree, Random Forest, and XGBoost, selecting the best-performing one based on evaluation metrics.

Outcomes:

Achieved high prediction accuracy, with XGBoost Regressor performing the best with an R^2 Score of 0.8883.

Enabled users to estimate car values accurately, supporting data-driven decision-making for buying and selling.

Strengthened my expertise in Python, Pandas, Scikit-learn, and data visualization techniques.

Tech Stack:
Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, XGBoost

Student Performance - Exploratory Data Analysis (EDA)

Key Skills:

Python | Feature Selection | Feature Extraction | seaborn | Matplotlib | Pandas | NumPy | data prepr | Data Processing | Data Cleaning | data | EDA | Data Visualization | Enco | Data Anlysis

Project Link: https://github.com/prerna-patil26/student-performance-eda

This project delves into the analysis of student performance, focusing on factors that influence academic outcomes. The dataset explores various aspects such as demographics, study habits, online learning participation, and lifestyle choices, including stress levels, sleep patterns, and social media usage. Through Exploratory Data Analysis (EDA), we identify key trends and relationships that help in understanding the drivers of student success. The goal is to provide valuable insights that can guide educators, parents, and policymakers in fostering academic improvement.

My Role: As the lead developer for this project, I was responsible for the complete data analysis process, including:

Data Collection & Cleaning – Gathered the dataset and performed necessary data cleaning steps such as handling missing values and performing feature engineering to ensure the data is ready for analysis.

Exploratory Data Analysis (EDA) – Leveraged visualizations using tools like Matplotlib and Seaborn to explore trends, correlations, and key insights in the data. Various plots, including boxplots, bar charts, and heatmaps, were used to understand factors such as study habits, stress levels, and exam performance.

Insight Extraction & Reporting – Analyzed the relationships between student performance and various factors like study hours, learning styles, and lifestyle habits. Key findings were summarized and presented for actionable insights.

Outcomes:

Uncovered actionable insights such as the positive impact of consistent study hours and the negative effects of excessive social media use on academic performance.

Highlighted the significance of sleep patterns and stress management for improving student outcomes.

Generated insights that can help educators create tailored strategies for enhancing student performance.

Tech Stack: Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn

Blinkit Grocery Data Analysis

Key Skills: SQL MySQL Power BI Microsoft Excel Visualization

Project Link: https://github.com/prerna-patil26/Blinkit-Grocery-Analysis

Project Overview:
This project analyzes Blinkit grocery sales data to gain insights into customer behavior, product demand, and delivery trends. Using Power BI and Excel, it provides a detailed breakdown of sales patterns, payment methods, and delivery performance to support business decision-making.

My Role:
• Created Pivot Tables and Charts in Excel to visualize key metrics such as sales trends, revenue distribution, and product performance.

• Developed Power BI dashboards with interactive charts, slicers, and filters to analyze data efficiently.

• Explored order trends, payment preferences, and delivery types to identify patterns and insights.

Key Outcomes:
• Identified top-selling products and high-revenue categories for better sales strategies.

• Analyzed delivery trends to understand performance across different cities and order types.

• Improved data visualization skills using Excel and Power BI.

Tech Stack:
Power BI, Excel, Pivot Tables, Data Visualization.

SEMINARS / TRAININGS / WORKSHOPS

Business Intelligence Using Advanced Excel & Power BI

27 Feb, 2025 - 20 Apr, 2025

Institute Name: ExcelR EdTech Pvt. Ltd.

Key Skills: Microsoft Excel Power BI

The "Business Intelligence Using Advanced Excel & Power BI" course equipped me with advanced skills in data analysis, visualization, and reporting. I learned how to use Excel's advanced functions (like pivot tables, Power Query), create interactive dashboards in Power BI, and apply DAX formulas for data modeling and advanced calculations to derive meaningful insights. Key Skills Learned: Advanced Excel techniques (formulas, pivot tables, Power Query). Power BI for data modeling, creating interactive dashboards, and applying DAX formulas. Data cleaning, transformation, and visualization for impactful reporting.

Outcome: Improved my ability to analyze complex data and present insights through reports, dashboards, and advanced calculations using DAX in Power BI.

PERSONAL INTERESTS / HOBBIES

- travelling , reading , learning new things

WEB LINKS

- Github - https://github.com/prerna-patil26

PERSONAL DETAILS

Gender: Female
Marital Status: Single
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